

Nifty 100 Analytics Analyst Guide

A practical guide to the screener, dashboard, PDF reports, and REST API built across Sprints 1-6 of the Nifty 100 Analytics project.

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1. Project Overview

Nifty 100 Analytics is an end-to-end equity research toolkit covering all 92 companies with usable data in the Nifty 100 index. It ingests 12 source Excel files into a single SQLite database (nifty100.db), computes 45+ financial ratios and CAGR metrics per company-year, runs a rule-based screener and peer percentile engine, generates PDF tearsheets and sector reports, produces auto-generated pros/cons via an NLP layer, clusters companies into 5 archetypes, and exposes everything through a Streamlit dashboard and a FastAPI REST API.

Key Makefile commands:

```
make load      - load all Excel files into nifty100.db
make ratios    - run the Ratio Engine, populate financial_ratios
make test      - run the full pytest suite, produce reports/pytest_report.html
make report    - generate all tearsheets, sector reports, portfolio report
make dashboard - launch Streamlit on localhost:8501
make api       - launch FastAPI on localhost:8000
make clean     - remove cache/test artifacts (database untouched)
```

2. Using the Streamlit Screener

Open the Screener page from the sidebar. Ten sliders on the left control the active filters: ROE min, D/E max, FCF min, Revenue CAGR min, PAT CAGR min, OPM min, P/E max, P/B max, Dividend Yield min, and ICR min. The results table updates live as you move any slider, and a result-count label above the table tells you how many companies currently match.

Six preset buttons — Quality Compounder, Value Pick, Growth Accelerator, Dividend Champion, Debt-Free Blue Chip, and Turnaround Watch — auto-fill the sliders to a curated combination. Note: the D/E filter automatically exempts Financials-sector companies (banks, NBFCs, insurers), since high leverage is structurally normal for that sector. The ICR filter treats a debt-free company (interest expense = 0) as having infinite interest coverage, so it always passes any ICR minimum.

Use the CSV download button beneath the results table to export exactly the columns currently shown, for further analysis in Excel or your own model.

3. Navigating the Dashboard

Home

Six summary KPI tiles (avg ROE, median P/E, median D/E, total companies, median Revenue CAGR 5yr, debt-free company count), a sector breakdown donut chart, and a top-5 companies table by composite quality score. A year selector in the sidebar updates every metric on the page.

Company Profile

Search by name or ticker. Shows the company card, 6 KPI tiles, a 10-year Revenue/Net Profit bar chart, an ROE/ROCE dual-axis line chart, and auto-generated pros/cons badges.

Screener

See Section 2.

Peer Comparison

Pick a peer group from the dropdown to see a radar chart (company vs peer average) and a side-by-side KPI table with the benchmark company highlighted.

Trend Analysis

Search a company and overlay up to 3 metrics on a 10-year line chart with YoY % change annotations.

Sector Analysis

A bubble chart (Revenue x ROE, sized by Market Cap) plus a sector median KPI bar chart.

Capital Allocation Map

A treemap of all 92 companies grouped by their 8 capital allocation patterns (Reinvestor, Shareholder Returns, Distress Signal, etc.) — click a pattern to see its company list.

Annual Reports

Search a company to see its available annual report years with clickable BSE PDF links; unavailable links show a red 'Report unavailable' badge.

4. Generating PDF Tearsheets and Reports

Run `make report` to regenerate everything: 91-92 two-page company tearsheets (skipping any company with fewer than 3 years of P&L; data — see `output/skipped_tearsheets.csv` for the current skip list), one PDF per broad sector, and the portfolio summary PDF (one page per company, alphabetical, with trend arrows).

Each tearsheet's page 1 covers 6 KPI tiles, a 10-year Revenue/Net Profit chart, and an ROE/ROCE dual-axis chart. Page 2 covers balance sheet composition, a cash flow waterfall for the latest year, auto-generated pros/cons, and a capital allocation pattern badge.

Individual tearsheets can also be regenerated for specific tickers only:

```
python3 -m src.reports.tearsheet TCS RELIANCE HDFCBANK
```

5. Calling the REST API

Start the API with `make api` (uvicorn on localhost:8000). Full interactive documentation is available at <http://localhost:8000/docs> (Swagger UI), and the raw spec is exported to `docs/openapi.json` (a ready-to-import Postman collection is also at `docs/postman_collection.json`).

Example requests:

```
curl http://localhost:8000/api/v1/health
curl http://localhost:8000/api/v1/companies/TCS
curl "http://localhost:8000/api/v1/screener?min_roe=15&max_de=1"
curl http://localhost:8000/api/v1/sectors/Information%20Technology/companies
curl http://localhost:8000/api/v1/peers/IT%20Services
curl -o tcs_tearsheet.pdf http://localhost:8000/api/v1/companies/TCS/tearsheet
```

All endpoints are read-only (GET). Unknown tickers/sectors/peer groups return HTTP 404 with a descriptive detail message; invalid screener parameter values return HTTP 400.

6. Understanding the KPIs

KPI	Meaning	Edge case handling
ROE	Net profit / (equity + reserves)	None if net worth <= 0
ROCE	(Operating profit + other income) / capital employed	None if capital employed <= 0
D/E	Borrowings / (equity + reserves)	0 (not None) if debt-free
ICR	(Operating profit + other income) / interest	'Debt Free' label if interest = 0
CAGR (Rev/PAT/EPS)	$((\text{end/start})^{1/n} - 1) \times 100$	6 edge cases: turnaround, decline-to-loss, both-negative, z
Capital Allocation	8-pattern classifier on sign of (CFO, CFI, CFF)	Reinvestor, Shareholder Returns, Distress Signal, etc.
Composite Quality Score	Weighted blend of ROE/ROCE/NPM/growth/leverage, 0-100	Sector-relative, winsorized version

7. Data Quality Notes and Known Anomalies

A small number of companies (BEL, HDFCLIFE, HAL, INDIGO) show ROE/ROCE values far above what's plausible, traced to a balance-sheet scale inconsistency in the source data rather than a formula bug — the same formulas match the reference file within 0.01 for the other ~85+ companies. These are documented, categorized, and flagged in `output/ratio_edge_cases.log`; the Sprint 6 clustering and outlier-detection modules winsorize/clip extreme values so a handful of data points don't distort cluster assignment or Z-score outlier flags.

17 companies (e.g. Asian Paints, Apollo Hospitals) have a `company_name` field in `companies.xlsx` that carries an embedded newline followed by descriptive text — all report/dashboard code truncates to the text before the first newline for display.

`stock_prices` and `market_cap` are SIMULATED datasets (clearly labeled as such in the dashboard and reports) — do not use for real investment decisions.

8. Troubleshooting Common Issues

`make api` fails with ModuleNotFoundError

Run ``pip install -r requirements.txt`` first — fastapi, uvicorn, and pydantic are required and are not part of the base Python install.

Dashboard shows 'Ticker not found'

Check the ticker spelling matches `companies.company_id` exactly (all-caps, e.g. TCS not tcs).

A tearsheet is missing for a company

Check `output/skipped_tearsheets.csv` — companies with fewer than 3 years of P&L; data are skipped by design.

Screener returns 0 or too many companies

Check whether you're combining a very tight threshold (e.g. $ROE > 50\%$) with a loose one — thresholds are combined with AND, not OR.

`make test` shows fewer than 60 tests

Confirm pytest, fastapi, and httpx are installed — two test files (`test_loader.py`, `test_endpoints.py`) are skipped if those packages are missing.